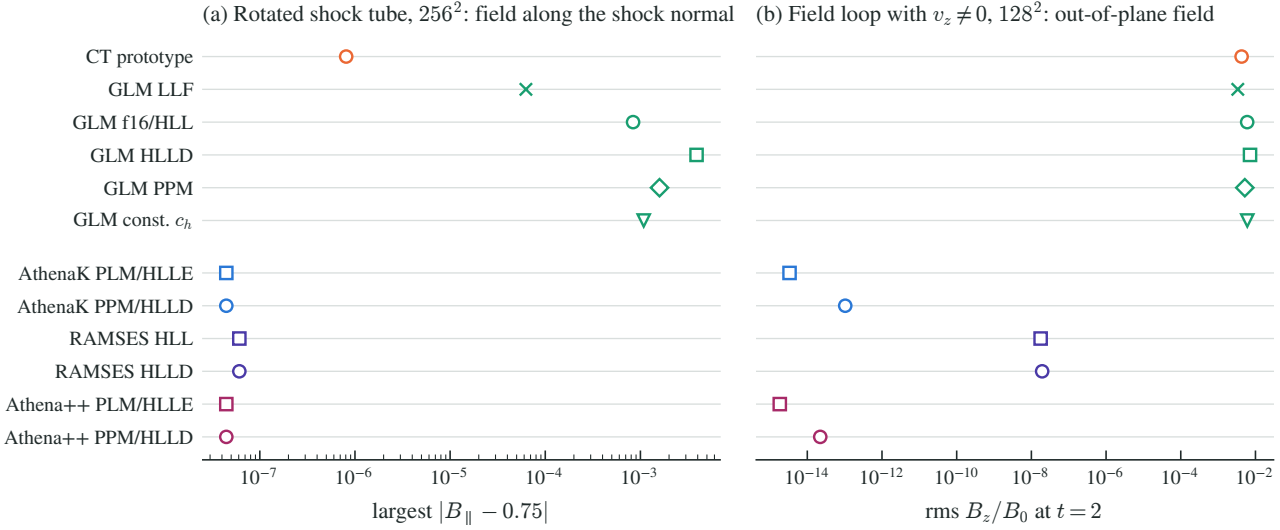
(a) Rotated shock tube, 256^2: field along the shock normal
(b) Field loop with $v_z \neq 0$, 128^2: out-of-plane field
CT prototype
GLM LLF
GLM f16/HLL
GLM HLLD
GLM PPM
GLM const. c_h
AthenaK PLM/HLLE
AthenaK PPM/HLLD
RAMSES HLL
RAMSES HLLD
Athena++ PLM/HLLE
Athena++ PPM/HLLD
10^{-7}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
largest $|B_\parallel - 0.75|$
10^{-14}
10^{-12}
10^{-10}
10^{-8}
10^{-6}
10^{-4}
10^{-2}
rms B_z/B_0 at $t = 2$